

HAT-P-12b

R-0761

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_240316 · created 2026-07-28T04:09:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460385.8519 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.054
Transit depth (Rp/Rs)^2	2.63 +/- 1.74 [%]
Orbital Inclination (inc)	86.73 +/- 1.67
Airmass coefficient 1 (a1)	0.971 +/- 0.044
Airmass coefficient 2 (a2)	0.04 +/- 0.19
Scatter in the residuals of the lightcurve fit is	4.61 %
Best Comparison Star	#3 - [213, 403]
Optimal Aperture	3.92
Optimal Annulus	7.44
Transit Duration (day)	0.073 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.162 $\pm$ 0.054 vs 0.1406 (deviation 0.4 σ)
Duration fit vs published	0.073 d vs 0.0975 d (deviation 0.84 σ)
Mid-time O–C	10.5 min
Depth SNR	3.0
Residual scatter	4.61 %

Light curve

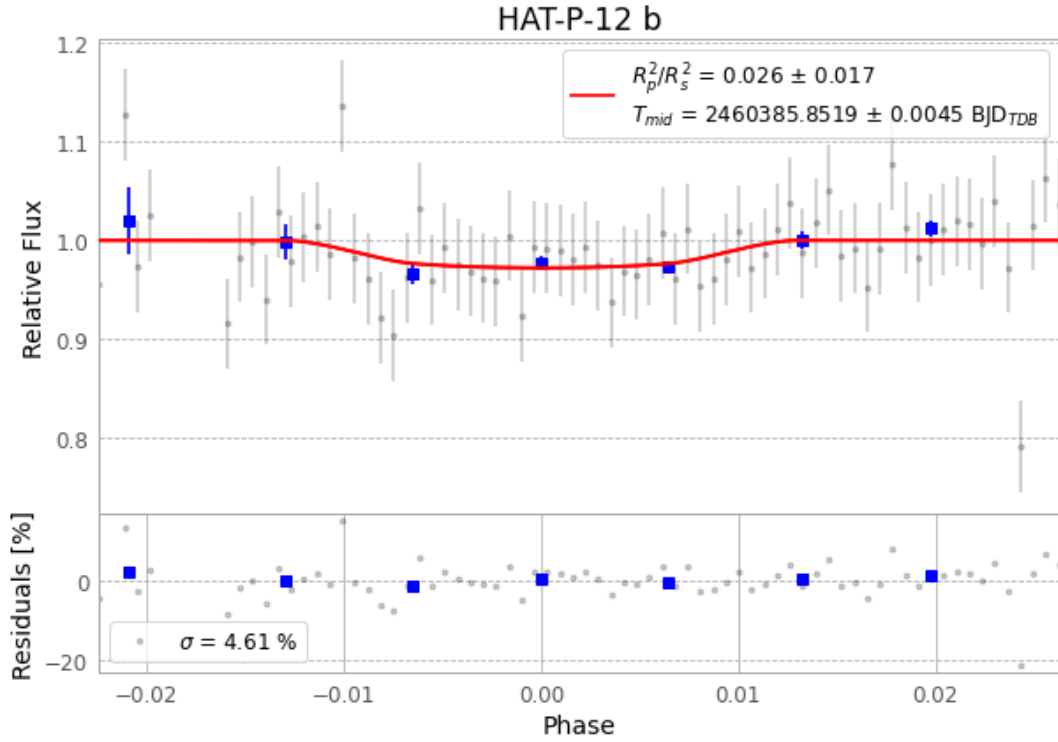


Figure 1 — FinalLightCurve_HAT-P-12 b_2024-03-15.png (SHA-256 d94b06c4c72134dc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8924255f3283eff1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), HATP-12240316060919.FITS → HATP-12240316102115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-240316.log (b9fd7a87c53a...), FinalLightCurve_HAT-P-12 b_2024-03-15.png (d94b06c4c721...), FinalLightCurve_HAT-P-12 b_2024-03-15.csv (18433066f5df...)

Compute resources

Wall time 230.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:09Z.